pfi

Python for Fixed Income

COURSE 10 · IMPLEMENTING RAG

Tuning the Pipeline: A Results Grid

WEEK 2 · DAY 3

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Today: a results grid

Twelve settings, six questions

Chunks	1,158 at 512/64, 520 at 1024/128
Retrieval	23 development questions, no request
Answers	6 development questions, saved runs
Requests	72 rows, 71 distinct
Held-out	not opened until day 11

Today

1. Names for settings
2. Expanding the context
3. What each setting sends
4. The grid, and reading it
5. Conservative sampling
6. Chunk size and depth together
7. Choosing on development questions only

Name every setting; write the rule before reading the table; six questions are few.

Twelve named settings

setting	what it changes in the base pipeline
k3	none: the base
k2	k=2
k5	k=5
chunk1024	size=1024, overlap=128
chunk1024_k2	size=1024, overlap=128, k=2
expand1	expand=1
hybrid	retrieve=hybrid
rerank	retrieve=hybrid, rerank=cross-encoder
temp03	temperature=0.3
temp07	temperature=0.7
top_p09	top_p=0.9
max_tokens150	max_tokens=150

temp0 is not a row: it is the base, one setting with two names and one hash.

A context window around the match

483

characters, the retrieved chunk

1111

characters, with one neighbour each side

3

chunks joined, overlaps removed

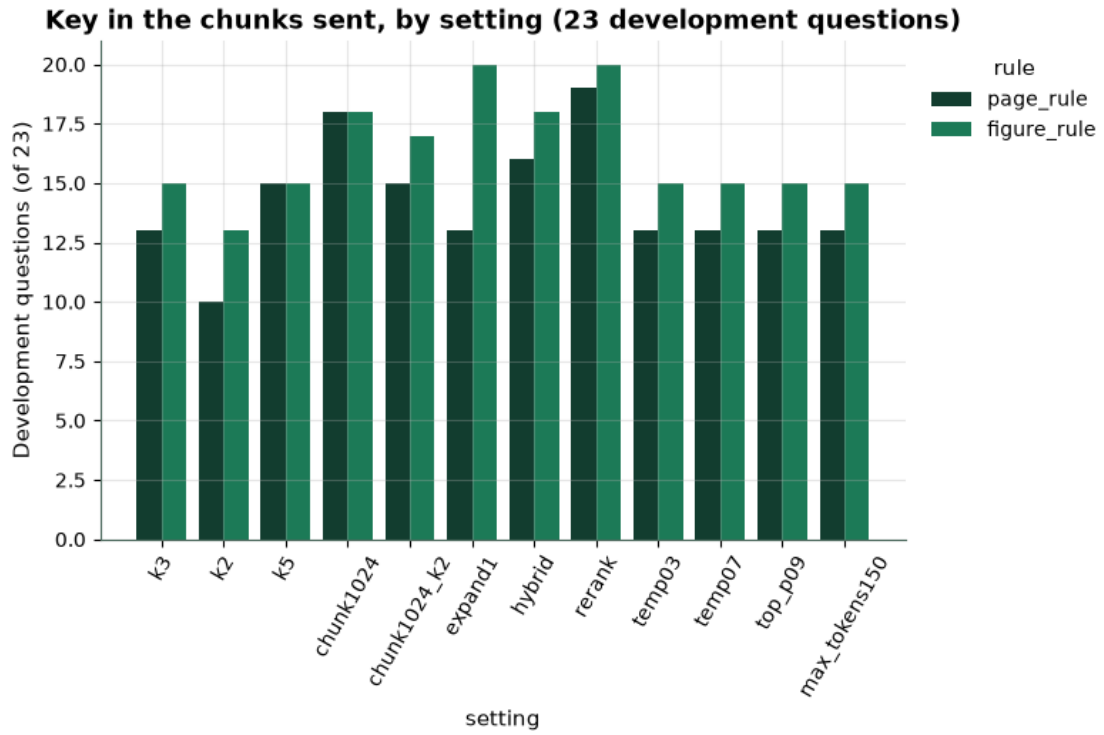
Chunks joined (dev13, top chunk)

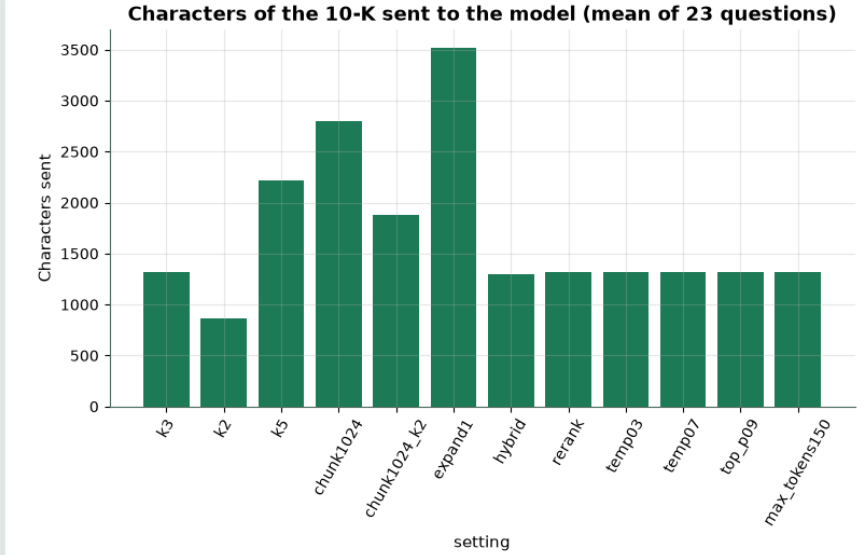
`boyd10k-8-059-0640, boyd10k-8-059-0641, boyd10k-8-059-0642`

The citation stays the retrieved chunk's; its neighbours are context.

Whether more text helps is a question for the grid, not for this cell.

What each setting sends



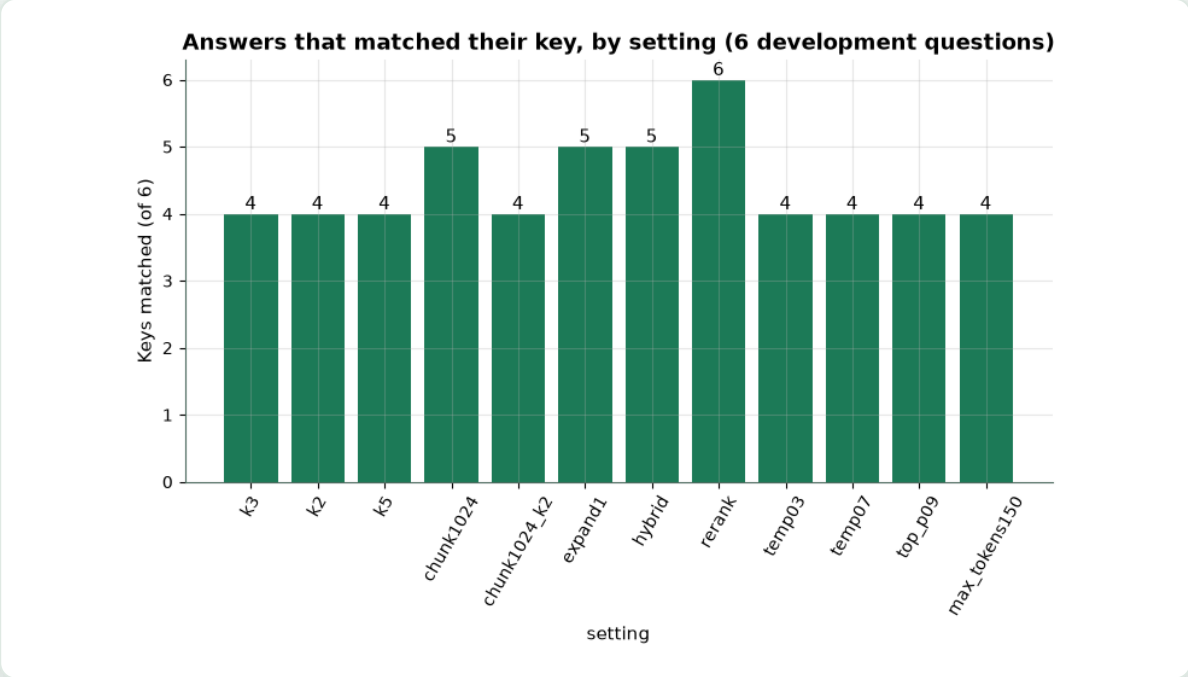


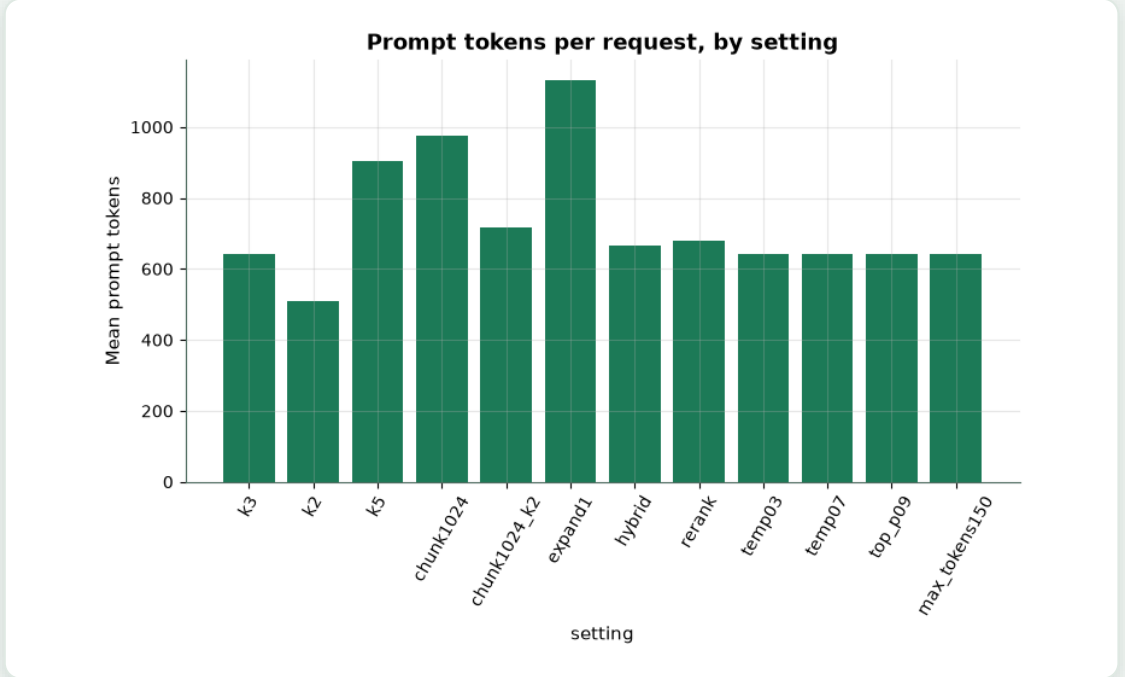
page rule: k3 13, k5 15, rerank 19

figure rule: k3 15, expand1 20

Of 23 development questions with a key page; no request is sent.

Twelve settings, six questions



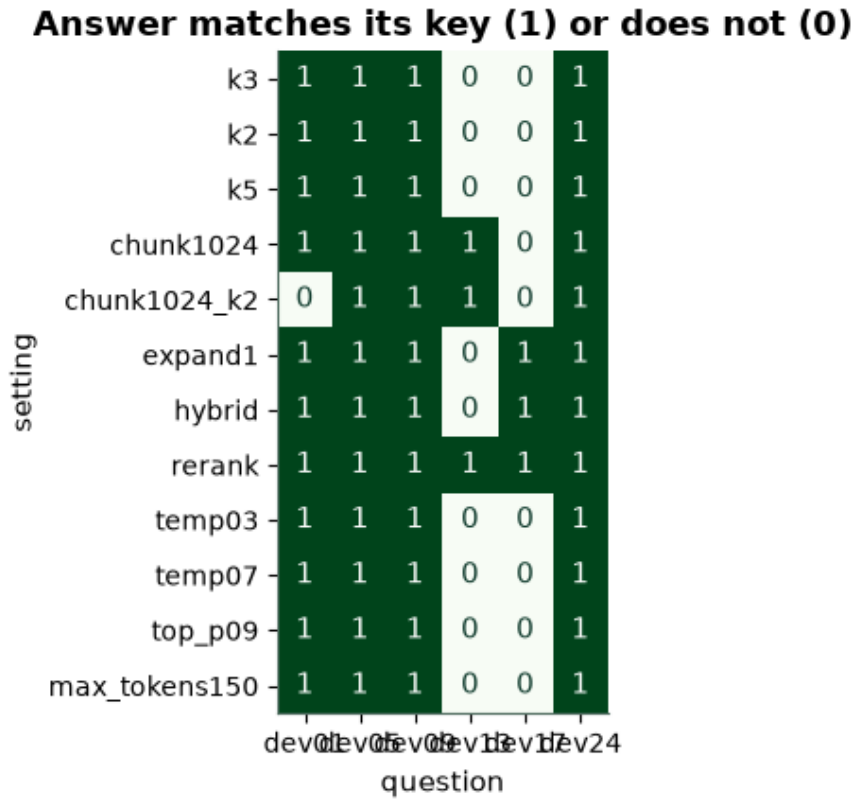


Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 72 requests, 52,746 prompt and 4,729 completion tokens; saved run recorded 2026-10-13; repla...

US$0.40 per million prompt tokens and US$1.60 per million completion tokens (developers.openai.com/api/docs/pricing, read 2026-10-13)

A count out of six moves one question at a time: one question is 17 points.

Which cells moved



Three readings

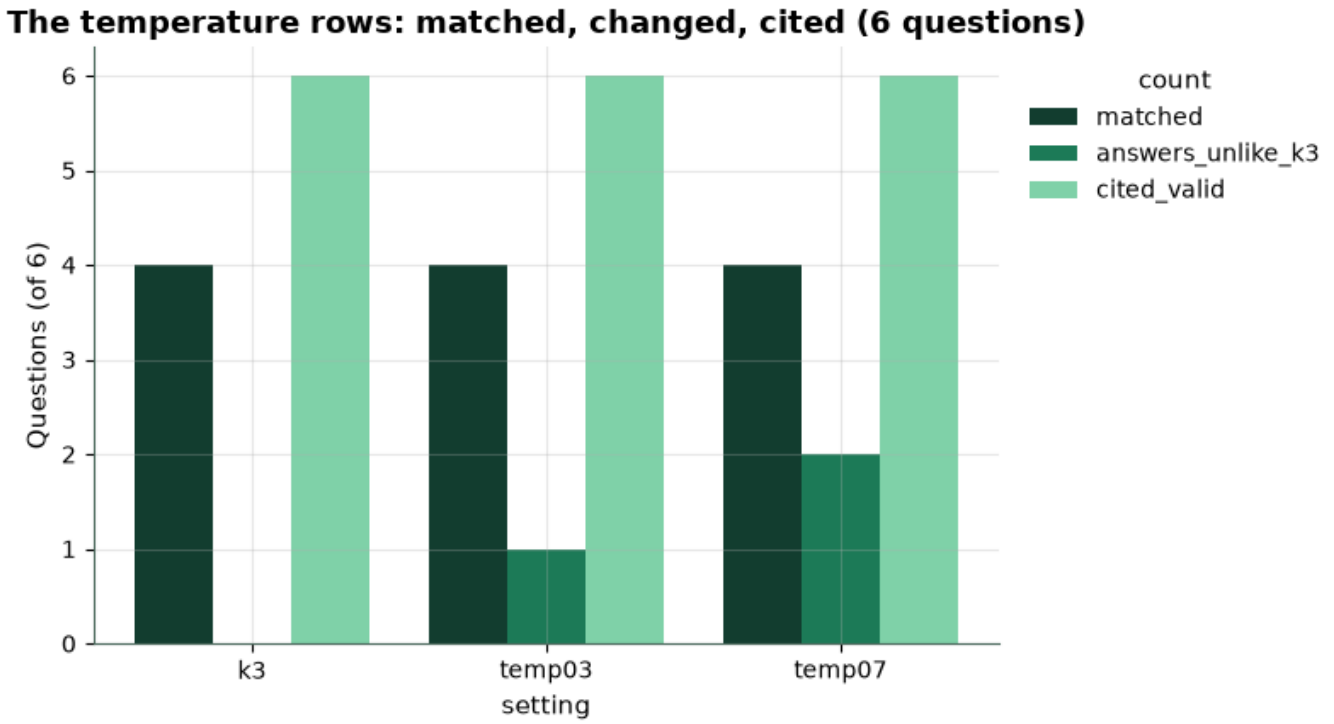
The settings differ on three of six questions: dev01, dev13, dev17. The other three do not depend on any setting here.

dev13: 8 settings answered 353,413 thousand, the right amount in the table's unit; the grader wants the key as written: no match.

Of 72 answers, 69 end every sentence with a citation; 23 hold the word will, quoted from the 10-K.

A changed text is not a changed grade; a grade measures a pipeline and a grader.

Conservative sampling, measured



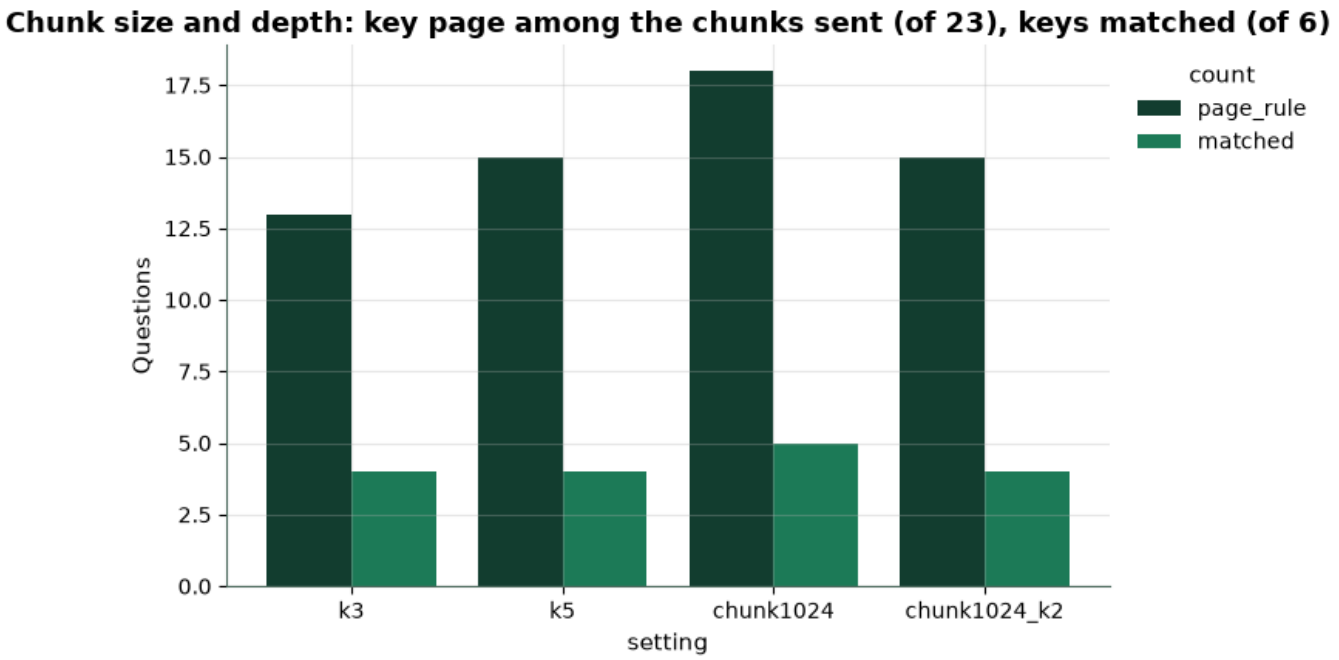
Six questions, one draw

temp	matched	unlike k3
0.0	4	0
0.3	4	1
0.7	4	2

No key was lost to the higher temperature. That is all six questions and one draw can say.

The SDK: alter temperature or top_p, but not both. The grid changes them in separate rows.

Chunk size and depth together



Characters sent, page rule, matched

k3 (512/64 at k = 3)

1,318 chars; page 13 of 23; matched 4 of 6

k5 (512/64 at k = 5)

2,214 chars; page 15 of 23; matched 4 of 6

chunk1024 (1024/128 at k = 3)

2,794 chars; page 18 of 23; matched 5 of 6

chunk1024_k2 (1024/128 at k = 2)

1,877 chars; page 15 of 23; matched 4 of 6

Liu and others, Lost in the Middle, arXiv 2307.03172 (2023-07-06; arXiv record read 13 October 2026). The paper itself was not read.

Choosing on development questions only

The rule, written first

1. Most keys matched, of 6

2. Tie: page rule on 23 questions

3. Tie: fewer prompt tokens

Development questions only

The twelve settings in the rule's order

setting	matched	page	tokens
rerank	6	19	678
chunk1024	5	18	975
hybrid	5	16	665
expand1	5	13	1133
chunk1024_k2	4	15	717
k5	4	15	905
k3	4	13	642
temp03	4	13	642
temp07	4	13	642
top_p09	4	13	642
max_tokens150	4	13	642
k2	4	10	509

Carried forward: rerank. The setting that matched the most keys on six development questions, not the optimal pipeline.

Excel side by side, and Git

Excel side by side (tested in Excel)

`=SUMIFS(runs[G],runs[A],A2)`	keys matched by rerank: 6, equal to Python
`=AVERAGEIFS(runs[G],runs[A],A2)`	share matched by k3: 0.6667
`=AVERAGEIFS(runs[G],runs[B],"512/64",runs[C],3)`	0.7500 over 48 rows
`=AVERAGEIFS(runs[H],runs[A],A2)`	mean prompt tokens, k3: 641.67
`=SORTBY(A2:A13,B2:B13,-1)`	the settings by keys matched: the same order

Git: rag_results.csv with 12 rows in two commits; 44 tests.

The AI check

The docstring is the prompt. The retrieval code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the filter keeps every question with a key page, so other splits get into the grid.

Code that runs is not the same as code that is right.

What you can do now

- Name every setting; differ from the base in one thing
- Count retrieval before paying for answers; name the rule
- run_grid refuses other splits and names the missing setting
- A changed text is not a changed grade
- Write the rule first; choose on development questions only

Week 2, Day 4 Metadata filters and many documents